

A unique natural human IgG antibody with anti-alpha-galactosyl specificity

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A new natural anti-alpha-galactosyl IgG antibody (anti-Gal) was found to be present in high titer in the serum of every normal individual studied. The antibody was isolated by affinity chromatography on a melibiose-Sepharose column. The reactivity of the antibody was assessed by its interaction with alpha-galactosyl residues on rabbit erythrocytes (RabRBC). The specificity was determined by inhibition experiments with various carbohydrates. The anti-Gal interacts with alpha-galactosyl residues, possibly on glycolipids of human RBC (HuRBC), after removal of membrane proteins by treatment with pronase. In addition, the anti-Gal bind specifically to normal and pathologically senescent HuRBC, suggesting a physiological role for this natural antibody in the aging of RBC. The ubiquitous presence of anti-Gal in high titers throughout life implies a constant antigenic stimulation. In addition to the theoretical interest in the antibody, the study of the anti-Gal reactivity seems to bear immunodiagnostic significance. Decrease in the antibody titer was found to reflect humoral immunodeficiency disorders.